

Sound detectors

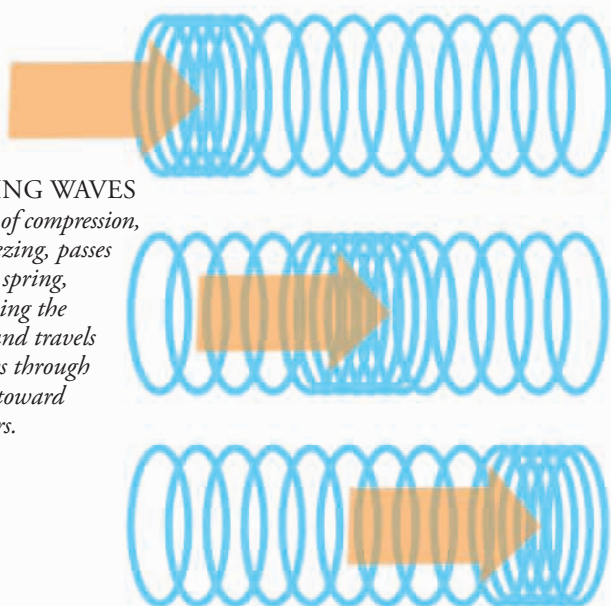
When you make a noise you send sound waves through the air. These are collected by your outer ears, which funnel the vibrations inward. Signals are sent from your ears to your brain that let you hear sounds and communicate using speech.

WAVES OF SOUND

Sounds are created by something vibrating and travel through the air as waves. How does this happen? Imagine pushing one end of a spring. Coils at the pushed end get squeezed then push apart, squeezing coils farther along, creating a wave of compression. With sounds, vibrations press air molecules together, causing waves to ripple away from the source.

► SPRING WAVES

A wave of compression, or squeezing, passes along a spring, mimicking the way sound travels as waves through the air toward your ears.



The ear flap (pinna) helps direct sound waves into the outer ear canal.

Cartilage provides springy support for the flexible earflap.

The ear lobe is filled with fatty tissue.

HOW YOUR EARS WORK

Sound waves enter your ear along its outer canal and hit the thin, skinlike eardrum, making it vibrate. These vibrations are passed to three tiny bones, called ossicles, that cross the middle ear. They move back and forth, pushing and pulling at the oval window, a membrane at the entrance to your inner ear. This sends ripples through fluid in the coiled tubes of the snail-shaped cochlea. Tiny hairlike receptors in the cochlea turn the ripples into nerve signals that are sent to the brain.

▲ INSIDE YOUR EAR

Each ear has three sections—the outer, middle, and inner ear. The outer ear canal is separated from the air-filled middle ear by the eardrum. The fluid-filled inner ear is entered through the oval window and contains the spiral cochlea.

WOW!

Many mammals, including porpoises, bats, and cats, can hear very high-pitched sounds, which humans cannot detect.